

Skin Cancer Identification utilizing CNN with PSO-driven Feature Selection and ML Classifiers

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Declaration

We hereby declare that the Mini - Project(A8041) titled "**Skin Cancer Identification utilizing CNN with PSO-driven Feature Selection and ML Classifiers**", submitted to Vardhaman College of Engineering (Autonomous), affiliated with Jawaharlal Nehru Technological University Hyderabad (JNTUH), in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (AI & ML)**, is the result of original work carried out by us.

We further certify that this project report, either in full or in part, has not been previously submitted to any university or institute for the award of any degree or diploma.

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Abstract

Skin cancer is one of the most common and deadly diseases which exists worldwide. The process of early and accurate detection helps to increase patient survival rates. Traditional diagnostic methods require dermatology experts to perform their work which results in assessments that take a long time and show different results. The project introduces an Explainable Artificial Intelligence (XAI)-driven hybrid framework which uses deep learning and optimization techniques to automate skin cancer detection. The system uses Convolutional Neural Networks (CNNs) to extract essential features from dermoscopic images. The system uses Particle Swarm Optimization (PSO) to select features which produces better classification results while decreasing the amount of processing power needed. The selected features undergo further processing through machine learning classifiers which include Support Vector Machines (SVM) and Random Forest and Logistic Regression to enhance their ability to predict outcomes. The proposed model is evaluated on benchmark datasets such as HAM10000 or ISIC which show better results in accuracy and precision and recall and F1-score when compared to standard methods. This system uses deep learning and swarm intelligence and explainable AI to provide dermatologists with a trustworthy and efficient and understandable method which identifies skin cancer at an early stage.

Keywords: Explainable Artificial Intelligence (XAI), Skin Cancer Detection, Convolutional Neural Networks (CNN), Particle Swarm Optimization (PSO), Feature Selection, Machine Learning, Dermoscopic Image Analysis, Deep Learning, Grad-CAM, Medical Image Classification, Early Diagnosis, Clinical Decision Support

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Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
PSO	Particle Swarm Optimization
XAI	Explainable Artificial Intelligence
SVM	Support Vector Machine
RF	Random Forest
Grad-CAM	Gradient-weighted Class Activation Mapping
ISIC	International Skin Imaging Collaboration
HAM10000	Human Against Machine with 10000 Images Dataset
UI	User Interface
TP	True Positive
FP	False Positive
TN	True Negative
FN	False Negative

CHAPTER 1

Introduction

1.1 Introduction

The healthcare industry has started to implement artificial intelligence systems for patient treatment and disease identification in recent times. Skin cancer has become one of the deadliest and fastest increasing cancer types that affect people throughout the world. The survival rates for melanoma patients show significant improvement when doctors detect the disease at its earliest stage. The diagnostic process in traditional methods relies on dermatologists' knowledge, which results in both diagnostic delays and inconsistent results.

The development of Artificial Intelligence (AI) technologies together with Machine Learning (ML) and Deep Learning (DL) technologies has made automatic skin lesion assessment possible. The systems analyze dermoscopic images to extract advanced features and achieve precise lesion classification. Deep learning systems operate as "black boxes" because they do not show how they reach their decisions.

The project introduces an Explainable AI-based skin cancer detection system which employs CNN for feature extraction and PSO for feature optimization and machine learning classifiers to achieve precise predictions. The system uses visualization methods, including Grad-CAM, which enables users to understand its operations, thus establishing its trustworthiness for practical medical use.

1.2 Proposed System Architecture

The proposed system is designed as a hybrid intelligent pipeline that combines deep learning, feature optimization, and classical machine learning techniques to achieve accurate skin cancer classification. Instead of relying on a single model, the architecture integrates multiple components, each contributing to improved performance and

interpretability.

1.2.1 Architecture Overview

The system begins with dermoscopic image input and processes it through several stages including preprocessing, feature extraction, feature selection, classification, and explainability. The overall workflow is illustrated in Figure 1.1.

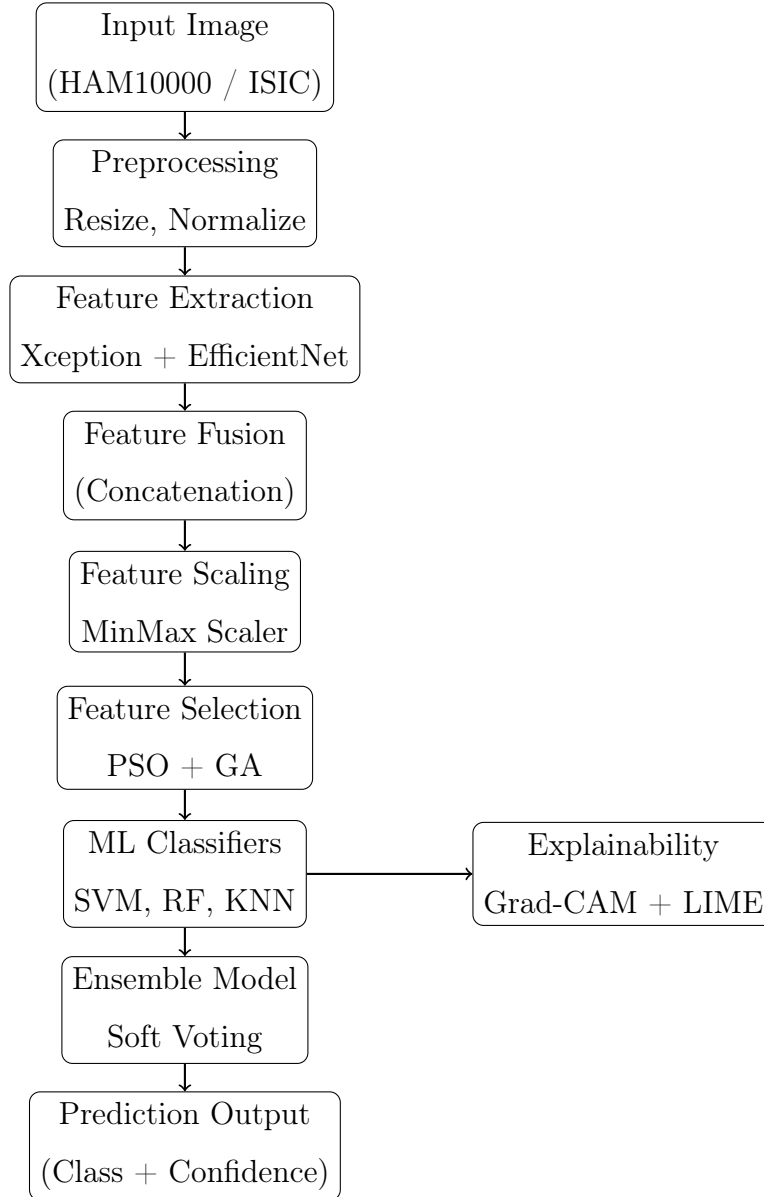


Figure 1.1: Overall Architecture of the Proposed Skin Cancer Detection System

1.2.2 Detailed Workflow Description

1. Input Layer: The system accepts dermoscopic images collected from publicly available datasets such as HAM10000 and ISIC. These images represent different

categories of skin lesions.

2. Preprocessing: Each image is resized to a fixed dimension of 224×224 pixels and normalized to ensure consistency. This step helps in reducing noise and preparing the data for deep learning models.

3. Feature Extraction: Two pre-trained convolutional neural networks, namely Xception and EfficientNetB0, are used to extract deep features from the images. These models are not retrained but used as powerful feature extractors.

4. Feature Fusion: The features obtained from both CNN models are combined into a single feature vector. This fusion helps in capturing complementary information from different architectures.

5. Feature Scaling: A MinMax scaler is applied to normalize the feature values between 0 and 1. This improves the performance of machine learning classifiers.

6. Feature Selection (PSO + GA): To reduce redundancy and improve efficiency, a hybrid optimization approach combining Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) is used. This step selects the most relevant features from the fused feature set.

7. Classification: The selected features are fed into multiple machine learning classifiers, including Support Vector Machine (SVM), Random Forest (RF), and K-Nearest Neighbors (KNN). Each model independently predicts the class of the lesion.

8. Ensemble Learning: A soft voting ensemble combines the predictions from all classifiers. The final decision is made based on the average probability, leading to improved accuracy and robustness.

9. Explainability: To make the system more transparent, Grad-CAM is used to visualize important regions in the image, while LIME explains the decision of the ensemble model. This enhances trust and interpretability.

10. Output: Finally, the system outputs the predicted class of the skin lesion along with a confidence score, helping in better clinical understanding.

1.2.3 Key Advantages of the Architecture

- Combines deep learning and machine learning for improved performance.
- Reduces feature dimensionality using intelligent optimization.
- Enhances accuracy through ensemble learning.
- Provides interpretability using Grad-CAM and LIME.
- Efficient and suitable for real-time prediction.

1.3 Background and Motivation

Skin cancer is one of the most common types of cancer in the world, and melanoma is the most dangerous form because it has a high potential to spread rapidly. Thus, early detection of skin cancer increases survival probabilities and reduces treatment difficulties. Dermatologists have traditionally used visual inspection and dermoscopic analysis to detect skin cancer. However, this method depends on the expertise of the dermatologist and may sometimes result in misdiagnosis, especially in early stages.

The significant progress in Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) techniques has enabled the development of automatic systems to assist in medical image analysis. Convolutional Neural Networks (CNNs) have been found to have outstanding potential in feature analysis in dermoscopic images and skin lesion classification with high accuracy. These systems have the ability to identify complex features that cannot be easily detected by humans.

The application of Deep learning models is effective but cannot be explained because people consider it a black box system. The application in medical science faces a decline in trust among doctors because it operates without an explanation mechanism. Thus, it is not possible to trace how it reaches its final result. Explainable Artificial Intelligence (XAI) techniques are urgently required because they have the ability to display how it produces its results.

The extracted features obtained using Deep learning models include both required and non-required features, resulting in a duplicate feature set that reduces skin cancer detection accuracy. The most important features in a data set can be obtained using

Particle Swarm Optimization (PSO) because it acts as an optimization technique that improves system performance while reducing system resource usage.

The necessity of developing real-time diagnostic systems is required because users and doctors require an easy-to-use system that provides results in real-time. The system attains this feature through the integration of AI models with web application techniques.

1.4 Objectives of the Project Work

The specific objectives of this project are:

- i. In order to extract the necessary characteristics from the transactional data, such as the transaction amount, location, etc., that aid in distinguishing between legitimate and fraudulent transactions.
- ii. To develop a CNN model for feature extraction from skin images
- iii. To apply PSO for feature optimization
- iv. To use ML classifiers (SVM, RF) for classification
- v. To implement Grad-CAM for explainability
- vi. To deploy the system using Flask for real-time prediction
- vii. To evaluate performance using accuracy, precision, recall, F1-score

1.5 Organization of the Report

The field of medical image analysis uses Machine Learning (ML) and Deep Learning (DL) techniques for skin cancer detection because these technologies now play a critical role in this area. The system uses dermoscopic images to detect facial patterns which enables precise skin lesion identification through its automated learning process. The system uses AI-based technology to handle extensive image data through automatic analysis which creates constant results that do not require human input.

Machine learning and deep learning models use dermoscopic image analysis to study skin lesions through their color, texture, shape, and structural pattern characteristics. The Convolutional Neural Network (CNN) architecture provides excellent capabilities

for automatic complex feature extraction from high-level data. These models can identify subtle variations in skin lesions that are often difficult for human observation, thereby improving diagnostic accuracy.

The techniques demonstrate complete success because they maintain strong performance across different datasets. The models learn skin lesion patterns through their training on extensive datasets which include HAM10000 and ISIC and they achieve accurate predictions for unknown images. The traditional rule-based systems display less stability than AI systems because AI systems possess superior strength throughout various operational conditions.

The medical field requirements for deep learning models need the ability to interpret their results. The solution to this problem requires the implementation of Explainable Artificial Intelligence (XAI) solutions which use Grad-CAM to show visual explanations of how models forecast results. The techniques demonstrate which image areas have significance because they affect the decision-making process of the model. The techniques create greater confidence because they enable users to understand how decisions were made.

Chapter 1: Introduction This report presents the project overview together with its background information and motivation and objectives which focus on skin cancer detection through artificial intelligence methods. The document demonstrates that developing accurate and efficient and explainable models serves as the main research goal because these models will help with early diagnosis and clinical decision-making improvements.

Chapter 2: Literature Survey The second chapter of the text presents an in-depth examination of current research studies which exist in the domain of skin cancer detection. The study covers both traditional diagnostic techniques and the latest developments in Machine Learning and Deep Learning methods. The advantages and limitations of these methods are also discussed.

Chapter 3: System Design describes the overall architecture of the proposed system. The system includes image preprocessing and CNN-based feature extraction

and PSO-based feature selection and machine learning classifiers and explainability through Grad-CAM and Flask-based web application real-time prediction integration.

Chapter 4: Proposed Methodology The working of the proposed hybrid skin cancer detection model is explained in detail in this section. The process begins with image preprocessing which leads to CNN-based feature extraction followed by PSO-based feature optimization and then the model training process which uses machine learning classifiers and the system integrates explainability techniques that include Grad-CAM.

Chapter 5: Results and Discussion The experimental results together with the performance assessment of the skin cancer detection system are shown in this section. The system's effectiveness is tested through evaluation metrics which include accuracy, precision, recall, and F1-score to establish its capability to classify various skin lesion types.

Chapter 6: Conclusion and Future Scope The section presents an overview of the entire project which reveals all research findings. The research demonstrates how artificial intelligence methods benefit skin cancer detection while the study presents future updates which can improve the existing system.

Overall, the report offers a thorough understanding of the design, implementation, and evaluation of the intelligent fraud detection system using hybrid approaches of machine learning and deep learning techniques.

CHAPTER 2

Literature Survey

2.1 Introduction

The healthcare system faces its primary challenge in skin cancer detection because the number of global skin cancer cases continues to rise. The early diagnosis of skin cancer, particularly melanoma, is essential for effective treatment and improved survival rates. Doctors need automated diagnostic systems because the increased use of dermoscopic imaging and digital health systems requires help to attain precise diagnostic results which will assist them in their work while decreasing their diagnostic mistakes [10].

The chapter shows all present research work that investigates the detection of skin cancer through various methods. The researchers of this study developed three different approaches to skin cancer detection which include traditional diagnostic techniques and machine learning methods and deep learning systems. The study of these methodologies helps in understanding their efficiency, advantages, and limitations in detecting different types of skin lesions [9].

The traditional method for skin cancer detection uses dermatologists to conduct visual inspections and perform dermoscopic analysis. The methods show effectiveness because they need specialized clinician skills to complete their work. The approaches cannot identify early-stage abnormalities because they depend on detecting minor alterations in the characteristics of lesions [10].

The development of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) technologies has created advanced automated systems that can detect

skin cancer. Machine learning models can analyze image features such as color, texture, and shape, while deep learning models like Convolutional Neural Networks (CNNs) can automatically extract complex features and achieve high classification accuracy [10].

Researchers today work to create hybrid systems which merge deep learning methods with optimization approaches to achieve better results. The research uses Particle Swarm Optimization (PSO) for feature selection which results in increased model efficiency and decreased computational requirements [7]. The system uses Explainable Artificial Intelligence (XAI) methods which include Grad-CAM to create visual explanations of model predictions that enhance trustworthiness and transparency in medical contexts [6].

The field has made progress through developments yet it faces multiple issues which include medical dataset data imbalance problems and inconsistent image quality and requirements for instant systems that people can understand. The development of dependable and effective explanatory models should proceed because these models enable precise skin cancer identification [9].

The project aims to solve existing problems by creating a hybrid system which uses CNN for feature extraction and PSO for feature optimization and machine learning classifiers for precise prediction of skin cancer detection. The system uses explainability techniques which deliver transparent decision-making processes that build trustworthiness.

The literature review presented in this chapter helps identify research gaps and provides a strong foundation for the proposed system. proposed system

2.2 Review of Prior Research

2.2.1 Traditional Skin Cancer Detection Approaches

The earliest skin cancer detection methods were primarily based on clinical examination and dermoscopic analysis performed by dermatologists. The techniques involved skin lesion assessment through visual inspection which used established medical criteria that included asymmetry and border irregularity and color variation and diameter according to the ABCD rule. The traditional methods achieved widespread clinical use because of their basic design and straightforward results which created the first steps for diagnosing skin cancer. [10].

Traditional methods combined visual inspection with pattern recognition techniques that relied on manually designed features which included color and texture and shape elements that were drawn from dermoscopic images. The methods analyzed past medical records to find patterns which matched with fresh skin lesion photographs. The systems used current images to compare with earlier diagnosed cases which helped them identify typical skin condition patterns. The analysis and benchmarking process has relied on datasets such as HAM10000 and ISIC which have become standard resources for this purpose. [2].

The basic methods faced their first problem after showing their first success. The system needs expert knowledge for its operation because it requires manual process of feature extraction which creates irregularities in its diagnosis results. The early stages of cancer show skin lesions which contain minor differences that our eyes cannot easily perceive. The system fails to handle different types of image quality and lighting conditions and changes in lesion appearance.

The process of manually examining dermoscopic images requires a lengthy time commitment which becomes unmanageable when the total number skin cancer cases worldwide increases. The methods used in this research create a risk of misclassification because human mistakes can occur which results in postponed diagnosis and

treatment. The primary problem with conventional systems lies in their failure to perform effectively on new data that they have not encountered before.

Another limitation exists because traditional methods lack the ability to adapt to different situations. The methods do not acquire new knowledge nor develop their capabilities through time as modern AI systems do. The system's performance depends on its established rules and designer-created features, which limits its ability to handle complex skin lesion patterns that show diverse characteristics.

The existing difficulties which traditional skin cancer detection methods face today cannot cope with the advanced needs of contemporary medical imaging practices. The increasing availability of large-scale datasets such as HAM10000 and ISIC, along with advancements in Artificial Intelligence (AI), has led to the adoption of Machine Learning and Deep Learning techniques. The skin cancer detection system uses modern techniques to automatically extract image features and learn from fresh information while delivering accurate results and scalable and dependable solutions. [10,9].

2.2.2 Image-Based and Pattern Similarity Detection

Researchers have studied skin cancer detection through image-based methods and pattern similarity methods which go beyond standard diagnostic techniques by using dermoscopic image features for analysis instead of predefined clinical rules [cnn_skin]. The system analyzes visual features in skin lesions to match them against existing records of both benign and malignant skin conditions. The system uses feature assessment which includes color and texture and shape and structural pattern analysis to classify a lesion as cancerous or noncancerous [2].

Pattern-based methods show their best results when they need to identify skin appearance deviations from the standard. The case study shows that a lesion which presents irregular borders and unpredictable color patterns together with unusual texture characteristics should be treated as suspicious. The methods allow researchers

to find hidden early-stage abnormalities which would remain undetected through standard human examination methods. The methods use statistical measures and similarity metrics to analyze lesion image variations which include distance-based methods and clustering techniques [9].

The pattern similarity techniques use historical dermoscopic image datasets which include HAM10000 and ISIC to create reference patterns that they use to identify abnormal patterns. The systems operate by comparing new images against existing labeled data which enables them to detect minute changes that suggest possible malignancy. The system demonstrates greater flexibility through its ability to function with different skin lesion patterns as compared to traditional rule-based systems [2].

However, image-based and similarity methods also have certain limitations. The main problem arises because benign lesions display uncommon characteristics which create false positive results that resemble malignant tumors. The process of establishing precise similarity metrics for high-dimensional image data becomes challenging because different lighting conditions and resolution settings and image quality variations impact the system performance. The diversity of skin types and lesion appearances further increases the complexity of accurate classification [9].

The methods need substantial computational power because they must handle and analyze extensive collections of dermoscopic images during their operational period. As datasets such as HAM10000 and ISIC grow in size, scalability becomes an important challenge. The systems require optimization techniques like PSO to sustain operational efficiency during large-scale testing [7]

The study of image-based and pattern similarity methods advances intelligent skin cancer detection systems despite their existing limitations. The system establishes a fundamental basis for more advanced methods through its implementation of adaptive learning and anomaly detection features. The development of Machine Learning and Deep Learning-based systems emerged through the creation of CNN-based models

which automatically acquire complex visual patterns while processing high-dimensional data to enhance diagnostic accuracy throughout their operational lifetime. [10].

2.2.3 Machine Learning Approaches

The application of Machine Learning (ML) techniques has become increasingly important in the field of skin cancer detection, as these techniques can learn patterns from dermoscopic images and assist in accurate classification of skin lesions. Machine learning models demonstrate improved performance through their ability to adapt to changing image data patterns, which distinguishes them from conventional diagnostic techniques. [10].

Different machine learning algorithms such as Decision Trees, Random Forest, Support Vector Machines (SVM), and Logistic Regression have been widely applied for skin cancer classification. The algorithms use features derived from dermoscopic images which include color and texture and shape and structural patterns to identify between benign and malignant lesions. The HAM10000 and ISIC datasets serve as standard data sources for training and testing purposes of these models [2].

Researchers have developed hybrid machine learning models which merge multiple algorithms to achieve better classification results. Ensemble techniques use voting-based models together with boosting methods to improve output accuracy and system reliability through their multiple classifier capabilities. The methods help to boost generalization performance while decreasing the rate of incorrect classifications.

Machine learning methods encounter specific difficulties which need to be addressed. The systems need vast quantities of labeled data for their training process while their performance depends on the effectiveness of their feature extraction methods. The medical datasets in their current state show class imbalance problems because certain skin lesion types are not adequately represented. The limitations of machine learning techniques create challenges yet they establish a solid basis for building effective skin cancer detection systems [9].

2.2.4 Deep Learning Approaches

Deep learning functions as an essential component for detecting skin cancer because it automated learning processes, which enable systems to understand intricate and specialized patterns that exist within dermoscopic images. The medical image analysis field benefits from deep learning models because they process raw image data without requiring users to perform manual feature extraction, which distinguishes them from traditional machine learning approaches [10].

For this project, deep learning uses Convolutional Neural Networks (CNNs) which serve as the primary method for executing image classification tasks. The CNN system extracts spatial features from skin lesion images which include edges and textures and patterns to enable precise identification of different skin cancer types. CNN models demonstrate high accuracy and robustness for skin abnormality detection through their training on extensive datasets which include HAM10000 and ISIC. [2].

The current progress in skin cancer detection technologies uses deep learning together with optimization methods and explanation techniques for skin detection systems. The study uses Particle Swarm Optimization (PSO) as a feature selection method because it improves model performance while decreasing computation requirements. The system uses Explainable Artificial Intelligence (XAI) methods which include Grad-CAM to create visual explanations of model predictions that demonstrate crucial areas of skin lesions [6].

Modern research also investigates advanced deep learning architectures which include ResNet VGG and EfficientNet to enhance skin lesion classification performance. The models can capture more profound and intricate feature representations which results in improved diagnostic accuracy.

Deep learning models enable effective skin cancer detection because their system can

process extensive image data while identifying intricate patterns and their built-in classification methods achieve better results. The implementation of optimization and explainability methods with these models results in their enhanced suitability for actual clinical settings [10,9].

2.3 Identified Research Gaps

Skin cancer detection research has achieved major advances but three important gaps still limit the performance of existing detection systems which clinicians use in actual medical settings.

- i. **Limited Hybrid Model Optimization:** Current models demonstrate an inability to successfully combine deep learning techniques through CNN with Particle Swarm Optimization and other optimization methods. The creation of better hybrid frameworks which use advanced optimization methods to improve feature selection and model performance requires immediate attention.
- ii. **Imbalanced Dataset and Limited Feature Representation:** Datasets like HAM10000 and ISIC experience class imbalance problems because certain skin lesion types such as melanoma receive less representation. The model fails to detect rare critical cases because of the existing imbalance. The lack of proper feature optimization limits the system's ability to classify different elements.
- iii. **Dependence on Static Datasets:** Most existing systems rely on pre-collected dermoscopic image datasets and do not incorporate continuous learning from new data. The model cannot adapt to different skin lesion appearances because it depends on existing data which does not include ongoing research updates.
- iv. **Limited Real-Time Deployment:** Hospitals need better tools to deliver their services because existing web-based frameworks, which include Flask for deployment, do not support their real-time clinical operations. The healthcare sector currently struggles to manage substantial image data because it creates operational difficulties in real-world medical settings.
- v. **Basic User Interface and Visualization:** Many existing systems provide minimal user interaction and lack advanced visualization features. The existing

interfaces need improvements which should include detailed prediction insights and lesion highlighting features together with user-friendly dashboards to enhance their usability.

- vi. **Lack of Explainability:** Deep learning models function as black-box systems which prevent clinicians from establishing confidence in their predictions. The medical decision-making process lacks transparency and interpretability because there are no available explainable AI solutions which include Grad-CAM.
- vii. **Limited Evaluation Metrics and Clinical Validation:** Several studies focus primarily on accuracy without providing a comprehensive evaluation using metrics such as precision, recall, F1-score, and confusion matrix. The testing of clinical applications remains restricted because of limited access to both clinical validation and real-world verification methods. The testing of clinical applications remains restricted because of limited access to both clinical validation and real-world verification methods. The testing of clinical applications remains restricted because of limited access to both clinical validation and real-world verification methods.

2.4 Summary

The chapter presents a complete review of skin cancer detection methods which show how diagnostic practices developed from traditional methods to present-day intelligent systems. The conventional methods of skin cancer detection use visual inspection and dermoscopic analysis to assess skin conditions based on established clinical guidelines and the expertise of professionals. The methods offer basic functionality but their effectiveness falls short because they depend on human judgment and fail to identify minor skin lesion changes.

The proposed system uses advanced Machine Learning and Deep Learning methods to achieve precise skin lesion classification which addresses existing system constraints. Machine learning algorithms can learn patterns from dermoscopic image features such as color, texture, and shape which enables them to differentiate between benign and malignant lesions.

Deep learning uses machine learning methods and deep learning approaches that use Convolutional Neural Networks (CNNs) to perform automatic feature extraction. CNNs enable learning of intricate and advanced image patterns from dermoscopic images which results in better classification performance and eliminates the requirement of manual feature development. The models use HAM10000 and ISIC benchmark datasets to train on various skin lesion images which come with complete labels.

The proposed system uses Particle Swarm Optimization (PSO) to select features which will improve system performance. PSO helps in identifying the most relevant features, which decreases unnecessary features and increases model performance. The optimization process improves classification outcomes while decreasing processing demands.

A key strength of the proposed system is the integration of Explainable Artificial Intelligence (XAI) techniques, such as Grad-CAM, which provide visual explanations of model predictions. The skin lesion image analysis methods use these techniques to identify critical image areas, which helps medical professionals understand the results better.

The system functions as a real-time web application which uses the Flask framework for its implementation. The system allows users to upload dermoscopic images which create instant predictions because this feature makes the system usable in actual fieldwork.

The project demonstrates machine learning and deep learning model deployment through its practical application to real-world healthcare systems. The system begins with basic storage methods but develops into an efficient deployment system that can handle larger operations.

In conclusion, the proposed system that combines CNN PSO machine learning

classifiers and explainable AI techniques delivers an efficient solution which detects skin cancer with high accuracy while providing understandable results. The system solves traditional method limitations while it helps enhance early diagnosis and clinical decision-making processes.

CHAPTER 3

Prior Research and Methodologies

3.1 Introduction

The chapter provides an extensive evaluation of skin cancer detection methods because it focuses on both traditional diagnostic methods and machine learning and deep learning systems. Medical diagnosis systems require three main features for skin cancer detection purposes which include accurate results and understandable results and the ability to process information as it happens [10].

Traditional methods, which include visual inspection and dermoscopic analysis, need clinical rules that have been established beforehand. The methods fail to detect minor changes in skin lesions while they cannot handle automated analysis for extensive datasets. The automatic pattern detection in dermoscopic images requires machine learning and deep learning techniques as its primary method of operation [9].

The project employs Convolutional Neural Networks (CNNs) as the main deep learning model which will perform both feature extraction and classification tasks. The CNNs demonstrate strong capabilities to extract spatial features from dermoscopic images which include edges and textures and lesion structures. The convolution operation is mathematically expressed through the following equation:

$$f(x, y) = (I * K)(x, y) \quad (3.1)$$

[1,4,5,10,21,22]

where I represents the input image and K represents the convolution kernel

The application of Particle Swarm Optimization (PSO) for feature selection improves

model performance.

$$v_i^{t+1} = wv_i^t + c_1r_1(p_i - x_i^t) + c_2r_2(g - x_i^t) \quad (3.2)$$

$$x_i^{t+1} = x_i^t + v_i^{t+1} \quad (3.3)$$

Here, x_i denotes position, v_i velocity, p_i personal best, and g global best [7].

The optimized features are classified using machine learning algorithms such as Support Vector Machine and Random Forest [8,20].

The system uses Explainable Artificial Intelligence (XAI) methods such as Grad-CAM for interpretability [6].

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (3.4)$$

[9,16]

The performance is evaluated using precision, recall, and F1-score

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (3.5)$$

[9,16]

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3.6)$$

[9,16]

The evaluation of classification performance needs these metrics because they measure performance in datasets with imbalanced classes which exist in the HAM10000 and ISIC [2] datasets.

The system functions as a real-time web application which the Flask framework implements. Users can upload dermoscopic images and receive instant predictions. The system improves usability while enabling organizations to use it in actual healthcare

settings.

The proposed methodology uses CNN together with PSO and machine learning classifiers and explainable AI methods to create a complete solution which accurately and efficiently detects skin cancer while maintaining interpretability.

The machine learning algorithms Support Vector Machine (SVM) and Random Forest use the optimized features from PSO to perform classification tasks. The classifiers achieve better skin lesion classification accuracy because they learn to differentiate between various skin lesion types through their decision boundary learning process.

Explainable Artificial Intelligence (XAI) methods use Grad-CAM to improve system clarity through their implementation. The interpretation method uses Grad-CAM to create heatmaps which show vital sections of dermoscopic images that affect the model's prediction outcome [6].

3.2 Existing Work

3.2.1 Traditional and Rule-Based Approaches

The majority of early skin cancer detection systems were based on clinical examination and rule-based diagnostic criteria which needed expert knowledge to function. Dermatologists use the ABCD rule which assesses Asymmetry Border Color and Diameter to identify suspicious skin lesions. The systems operate through established guidelines which direct their decision-making process based on visible characteristics of dermoscopic images. The methods become popular in clinical practice because their interpretation allows users to comprehend them easily [10,19].

The primary reason for using rule-based methods during the first phase of automated skin cancer detection was their straightforward implementation process. The researchers used image processing and pattern recognition methods to create specific features which included color distribution and texture patterns and lesion shape from

dermoscopic images. The researchers used HAM10000 and ISIC datasets to compare these features with established patterns [2,3].

The traditional rule-based systems face multiple restrictions which prevent their complete operation. The primary limitation of the system occurs because it cannot handle changes in skin lesion appearance. Skin cancer exhibits multiple visual patterns which create challenges for predefined rules to establish universal rules. Simple rule-based methods fail to detect the minor distinctions which exist between benign and malignant lesions [10].

The system maintains two main types of errors which include false positive results that occur when noncancerous tumors display cancer-like characteristics and false negative results that happen when early-stage tumors lack visible detection signs. The inaccuracies in these tests lead to two problems which include errors in clinical decision processes and treatment delays [3].

The rule-based systems do not have the capability to learn from incoming data which they receive after their initial training. The systems from this era need human intervention to function better because they do not automatically enhance their performance throughout time. The system requires medical professionals to perform ongoing updates which limits its capacity to handle extensive data sets and operational use cases [9].

Traditional skin cancer detection methods now fail to deliver accurate results because of the existing limitations. The development of advanced Artificial Intelligence technologies together with existing large-scale datasets which include HAM10000 and ISIC has made Machine Learning and Deep Learning methods more effective for practical use. The systems enable automatic identification of intricate patterns in image data while handling different variations and delivering precise and extensive solutions for skin cancer assessment [10,2,3].

3.2.2 Machine Learning Approaches

The detection systems for skin cancer have experienced major advancements through Machine Learning (ML) techniques which enable systems to learn automatically from dermoscopic image data. The techniques enable the detection of intricate patterns which exist between features that include color and texture and shape and which standard methods are unable to identify [10].

Various machine learning algorithms such as Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines (SVM) have been widely used for classifying skin lesions. The models establish connections between extracted features and their respective output classes which include benign and malignant categories. The HAM10000 and ISIC datasets serve as standard training datasets that researchers use to test their models [10].

The combination of multiple classifiers through ensemble learning techniques which include Random Forest and boosting-based methods results in better performance because it decreases prediction errors. The methods improve skin cancer detection accuracy through their ability to enhance system reliability [20].

The field of machine learning methods shows particular drawbacks which exist as fundamental constraints. The methods need extensive labeled data and their performance relies on the quality of extracted features. Medical datasets face their primary challenge because certain lesion types get underrepresented in the data which results in model performance issues. The existing difficulties require implementation of sophisticated deep learning systems which can effectively handle the problem [9].

3.2.3 Deep Learning Approaches

Deep learning models have emerged as powerful tools for skin cancer detection because they can learn intricate nonlinear patterns which exist in dermoscopic images. Deep learning systems remove the requirement for human experts to create machine

learning system features that traditional methods need [10].

Convolutional Neural Networks (CNNs) are widely used in skin lesion classification tasks. The CNN system extracts spatial features from images which include edges and textures and patterns to achieve precise classification results. Models trained on benchmark datasets such as HAM10000 and ISIC have achieved dermatologist-level performance in some cases [10,2].

The performance of advanced deep learning architectures like ResNet and VGG and EfficientNet improves because these systems enable their users to extract more detailed features from their data and their models can better handle different types of tasks. The models recognized for their advanced visual processing capabilities can identify complex visual patterns found in skin lesions.

Deep learning models also support explainability through techniques such as Grad-CAM which shows crucial image parts that affect their prediction results in dermoscopic images. The system provides better transparency which boosts user confidence in AI systems used for medical purposes [6].

Deep learning models need extensive computational power together with large sets of labeled data because these requirements function as fundamental training conditions. The system demonstrates high efficiency in skin cancer detection because it automatically learns intricate patterns while handling various types of image data [9].

The combination of deep learning methods with Particle Swarm Optimization (PSO) and explainable AI techniques results in a strong and expandable system, which effectively detects skin cancer

3.2.4 Hybrid Approaches

Recent studies increasingly highlight the importance of adopting hybrid approaches that combine machine learning and deep learning techniques to enhance skin cancer

detection systems. The system uses multiple learning methods to detect both basic and advanced patterns present in dermoscopic images.

The Support Vector Machines (SVM) and Random Forest classifiers in traditional machine learning methods learn to separate different skin lesions through their training on labeled datasets which include HAM10000 and ISIC. The methods successfully classify the existing patterns that exist within the dataset. The system depends on manually extracted features which restrict its ability to handle new and complex image variations [2,3].

The other learning method uses deep learning models which include Convolutional Neural Networks (CNNs) to automatically extract important features from dermoscopic images. CNNs effectively recognize intricate visual patterns which include texture differences and lesion edges and color distribution patterns. The models contain multiple unnecessary features which decrease their operational efficiency [**cnn_skin**].

Hybrid models combine CNN-based feature extraction with Particle Swarm Optimization (PSO) optimization techniques to enhance their performance. The method uses CNN to extract deep image features while PSO selects important features through redundancy elimination which boosts classification performance [7].

The optimized features are then classified using machine learning algorithms which include SVM and Random Forest. The combination results in better classification accuracy while maintaining lower computational requirements. The system uses Explainable Artificial Intelligence (XAI) techniques which include Grad-CAM to create visual explanations that show critical areas of skin lesions which affect prediction results [6].

The system performance increases when the outputs from these components are effectively combined. The hybrid framework provides better accuracy and reliability and generalization ability than its separate models. The system uses two components

to decrease false positive rates while it boosts the reliability of medical diagnosis processes [9].

The study shows that hybrid systems handle changes in dermoscopic images better than other systems because they can adapt to different skin types and lighting conditions and various lesion types which are present in real-world clinical settings [2].

3.3 Comparative Analysis

Table 3.1: Comparison of Skin Cancer Detection Approaches

Ref	Methodology	Dataset	Technique	Accuracy
[1]	Xception CNN + PSO + ML (SVM, KNN) + XAI (Grad-CAM, LIME)	ISIC 2018, HAM10000	CNN + PSO + ML + XAI	98.5%, 86.1%
[2]	Active Learning + PSO + Transfer Learning	HAM10000	CNN + AL + PSO	~89.44%
[3]	CNN + PSO + Genetic Algorithm + SVM	Dermoscopic Images	CNN + Optimization + ML	89.17%
[4]	Multiple CNNs + Transfer Learning + ACO Optimization	HAM10000	CNN + ACO	96.5%
[5]	ConvNeXtV2 + Focal Self-Attention	ISIC 2019	Hybrid Deep Learning	93.60%
[6]	Systematic Review of ML, DL, Hybrid Models	ISIC, HAM10000, PH2	Survey Study	80–95%
[7]	Xception CNN + PSO + ML Classifiers	ISIC, HAM10000	CNN + PSO + ML	98.5%, 86.1%
[8]	CNN + PSO-based Feature Selection + ML	ISIC	CNN + Optimization	Improved

3.4 Summary

This chapter has provided a comprehensive analysis of various skin cancer detection techniques by discussing traditional machine learning deep learning and hybrid approaches. The traditional methods use visual inspection and dermoscopic analysis that follows predefined clinical rules such as the ABCD rule to provide straightforward solutions which people can understand. The methods fail to identify minor skin lesion changes because they require specialized knowledge from experts which results in inconsistent diagnosis outcomes [20, 22].

Machine learning techniques have made skin cancer detection more accurate through their ability to teach models which patterns to recognize in dermoscopic images and use that knowledge to evaluate new cases. The classification tasks which Random Forest and Support Vector Machines (SVM) and Logistic Regression algorithms performed with HAM10000 and ISIC datasets showed successfully results. Machine learning models face restrictions because they depend on manually created features which makes it difficult for them to interpret the intricate visual patterns found in medical images [9].

Deep learning methods enable advanced capabilities which automatically learn to process complex nonlinear patterns present in dermoscopic images. The combination of Convolutional Neural Networks (CNNs) with ResNet and EfficientNet advanced architectures shows high effectiveness for skin lesion classification. The models which exist can capture complex visual patterns which help to identify medical images more accurately [10,21,22].

The combination of CNN-based feature extraction with Particle Swarm Optimization (PSO) and other machine learning classifiers in hybrid approaches has resulted in better performance outcomes. The use of Explainable Artificial Intelligence (XAI) techniques which include Grad-CAM demonstrates skin lesion image analysis by showing vital regions which help boost confidence in model predictions[7,6].

The study demonstrates that traditional methods enable system interpretation but requires advanced machine learning and deep learning methods to create accurate and dependable systems for skin cancer detection. The system combines CNN, PSO, machine learning classifiers, and explainable AI methods to establish a robust base that enables development of diagnostic systems which maintain both efficiency and scalability and provide system understanding to users [10,9].

CHAPTER 4

Proposed Methodology

4.1 Introduction

The chapter provides a complete explanation of the skin cancer detection system methodology which uses advanced machine learning and deep learning methods to identify and classify various skin lesions. The system operates through its real-time functions which allow medical professionals to perform early patient diagnoses and treat patients based on their findings. The proposed methodology covers all stages of the system development life cycle [10].

The process begins with data collection, where dermoscopic image datasets such as HAM10000 and ISIC are used. The datasets contain labeled images which display different types of skin lesions that include both benign and malignant categories. The research needs suitable preprocessing techniques because medical datasets show imbalanced distribution patterns that decrease data quality [2,3].

The data preprocessing work needs three steps which require both image resizing and data normalization and noise removal and data augmentation. The steps in this process enhance the model's ability to generalize while improving its performance. The process of class imbalance correction represents a vital function which enables the model to acquire knowledge from all types of lesions

The first data preprocessing tasks need to finish their work because researchers need to conduct exploratory data analysis(EDA) which examines how the dataset distributes itself and shows its different attributes. The process of feature extraction uses Convolutional Neural Networks(CNNs) which learn to detect spatial features

through their automatic identification of edges and textures and the recognition of lesion patterns in dermoscopic images [10].

The application of Particle Swarm Optimization (PSO) for feature selection leads to better feature quality. PSO achieves its objective by optimizing the feature space and removing redundant features to identify essential features. The PSO update equations are given as:

$$v_i^{t+1} = wv_i^t + c_1r_1(p_i - x_i^t) + c_2r_2(g - x_i^t) \quad (4.1)$$

[7]

$$x_i^{t+1} = x_i^t + v_i^{t+1} \quad (4.2)$$

[7]

Here, x_i represents the position of the particle, v_i represents its velocity, p_i denotes the personal best position, and g denotes the global best position

Particle motion through the system displays their current location through x_i and their speed through v_i and their most recent achievement through p_i and the highest achievement of all particles through g [7].

The optimized features are then used for classification using machine learning algorithms which include Support Vector Machines (SVM) and Random Forest. The classifiers assist with enhancing both the accuracy and the robustness of skin lesion classification.

The system uses Explainable Artificial Intelligence (XAI) techniques which include Grad-CAM to make results more understandable. The model creates heatmaps through Grad-CAM which display the crucial areas of skin lesions that assist in predicting outcomes. The Grad-CAM formulation is given as:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (4.3)$$

[6]

Here, A^k represents the feature maps, and α_k^c represents the importance weights for class c

The equation shows that A^k stands for feature maps while α_k^c denotes importance weights [6].

The CNN system produces outputs which work together with PSO optimization results and machine learning classifiers to form a combined system. The system achieves better classification results because it reduces overfitting while enhancing its ability to generalize across different situations.

The system uses the Flask framework as its fundamental structure which enables users to test the prediction model through live interactions. Users can upload dermoscopic images and receive instant predictions which make the system useful for real-world healthcare settings.

The proposed methodology delivers a complete solution which enables the development of a skin cancer detection system that operates with high efficiency and precise accuracy while providing understandable results. The system applies deep learning combined with optimization techniques and explainable AI methods to solve existing limitations of common procedures which assist in early detection and support clinical decisions[9].

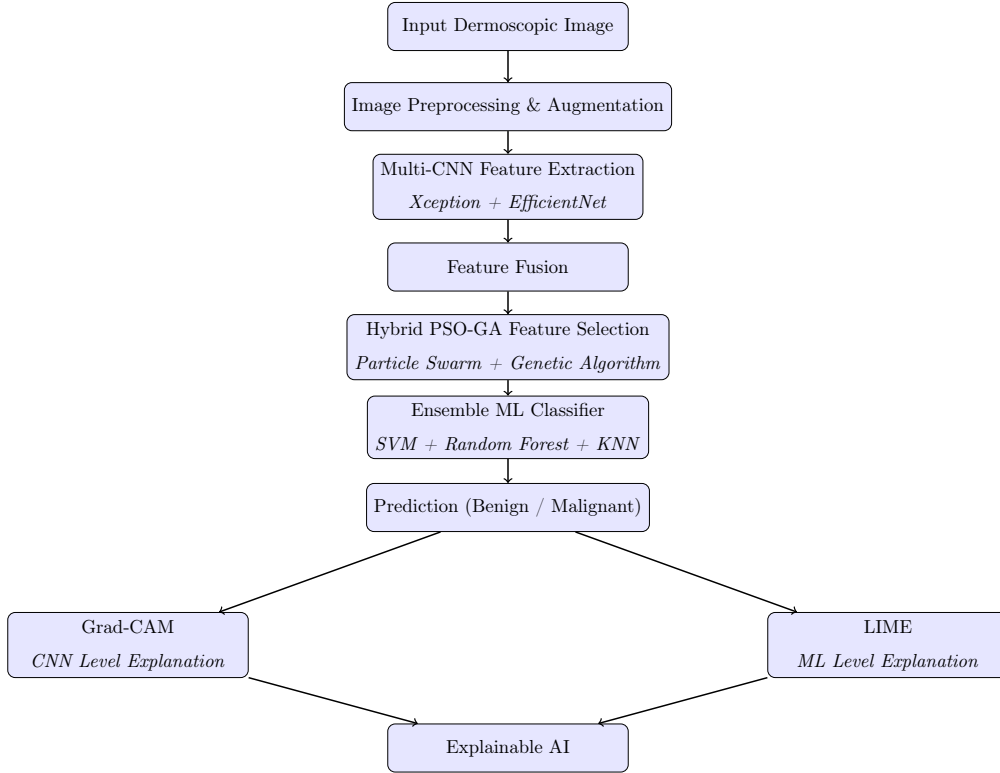


Figure 4.1: Proposed Explainable AI-Based Skin Cancer Detection Workflow

4.2 Dataset Description

The proposed system for skin cancer detection primarily utilizes benchmark dermoscopic image datasets such as HAM10000 and ISIC, which are widely used in medical image analysis research. The datasets provide high-quality images of different skin lesions, which researchers can use to develop and test automated skin cancer detection systems [2,3].

The HAM10000 dataset contains 10,000 dermoscopic images which researchers obtained from multiple populations to study different types of skin lesions including melanoma and nevus and benign keratosis. The ISIC dataset which contains extensive labeled skin lesion image data serves as a resource for researchers who want to conduct classification and segmentation research [2,3].

The dataset contains image data that includes dermoscopic images as its individual samples, which differs from structured transactional datasets. The dataset contains

several skin lesion types which create a multi-class classification challenge for researchers.

The datasets display class imbalance because some lesion types, including melanoma, occur less frequently than benign lesions. The model training process faces difficulties from this imbalance, which requires specific preprocessing methods. These methods include data augmentation and class balancing to enhance model performance [9].

4.2.1 HAM10000 and ISIC Dataset Features

The datasets include the following key attributes:

- i. **Dermoscopic Images:** High-resolution images of skin lesions used for feature extraction and classification.
- ii. **Lesion Categories:** Multiple classes such as melanoma, nevus, basal cell carcinoma, and benign keratosis.
- iii. **Metadata (Optional):** Additional information such as age, gender, and lesion location (available in some datasets).

4.2.2 Preprocessing of Image Data

Preprocessing steps begin before model training because they enhance data quality and model performance. The preprocessing process consists of these steps:

- Image resizing to a fixed dimension
- Normalization of pixel values
- Noise removal and enhancement
- Data augmentation (rotation, flipping, scaling)

The preprocessing methods improve model generalization while they decrease overfitting according to the research of [9].

4.2.3 Feature Representation

Feature extraction occurs through automated methods in Convolutional Neural Networks (CNNs) which operate differently from numerical datasets. CNNs learn to recognize hierarchical image structures that include edges and textures and patterns without requiring users to create specific features for their models [10].

4.2.4 Real-Time Image Input

The system provides live image processing through a web interface which functions together with its static dataset capabilities. Users can upload skin lesion images, which are processed by the trained model to generate predictions. The system uses this method to replicate actual user environments while it supports real-world implementation in medical settings.

Real-time image inputs require different handling than traditional datasets because they show actual user interactions which create dynamic content. The system can adapt to different input conditions because this feature enables it to learn from various situations.

The system achieves better operational performance through its real-time data functionality which lets it make instant predictions. This aspect holds special significance for medical fields because prompt diagnosis has the power to change patient results.

The system demonstrates its ability to operate in real-world environments because it can adapt to different image qualities and various lighting conditions and multiple skin types.

The proposed system demonstrates both practical implementation and effective skin cancer detection because it uses HAM10000 and ISIC datasets together with real-time image input.

4.3 Dataset Preprocessing

The proposed skin cancer detection system requires high-quality input data for its effective performance. The system uses a strong data preprocessing procedure which processes raw dermoscopic images to create structured data that can be used for training the model. The preprocessing step establishes the foundation for achieving model accuracy because it helps eliminate unwanted elements and solves the problem of unequal distribution of data throughout the dataset [2,9].

The following preprocessing steps are performed:

- i. **Data Cleaning and Validation:** The researchers analyze the HAM10000 and ISIC datasets to identify any images that are either damaged or of insufficient quality. The process eliminates all incomplete and noisy images which allows the researchers to maintain data integrity. High-quality reliable image data serves as the foundation for training the model according to this step [2].
- ii. **Duplicate Removal:** The process eliminates duplicate images from the dataset to avoid creating unnecessary multiple copies of the same content. The process decreases the chance of overfitting which allows the model to better generalize to new data [9].
- iii. **Image Resizing and Normalization:** All images are resized to a fixed dimension (e.g., 224×224 pixels) to ensure consistency. The process of normalization establishes a standard range for pixel value transformations which helps the model achieve faster convergence. The normalization is expressed as:

$$z = \frac{x - \mu}{\sigma} \tag{4.4}$$

[9,16]

Here, x represents the data point, μ is the mean, and σ is the standard deviation

The equation shows that pixel value x and mean value μ and standard deviation

σ together create the formula. The method guarantees that all features in the dataset will receive consistent scaling throughout the entire dataset [10].

- iv. **Handling Class Imbalance:** The dataset is imbalanced, as certain classes such as melanoma are underrepresented compared to benign lesions. The problem gets solved through the implementation of data augmentation, oversampling, and class weighting techniques, which enhance the performance of the model [3].

- v. **Data Augmentation:**

The researchers applied data augmentation techniques through three methods which included rotation and flipping and zooming and scaling. The model developed by the researchers gained better performance throughout different skin lesion image types because of these improvements [10].

- vi. **Feature Extraction and Selection:**

Convolutional Neural Networks (CNNs) execute feature extraction by automatically identifying crucial visual elements which include edges and textures and lesion patterns. Particle Swarm Optimization (PSO) is used for feature selection to keep only essential features while eliminating unnecessary features [7].

- vii. **Train/Test Split:**

The dataset is divided into training and testing sets, typically using an 80/20 split. The model evaluation process requires testing on new data which enables assessment of the model's ability to generalize from what it learned [9].

- viii. **Real-Time Image Preparation:**

The same preprocessing pipeline which processes static datasets also handles real-time input images which users upload through the web application. The system maintains uniformity between its training processes and deployment operations.

The preprocessing pipeline transforms dermoscopic image data into its required state for model training operations. The system improves data quality while decreasing background noise and achieving better results through its classification process.

The process of normalization and scaling establishes a standardized range for pixel values, which maintains stability during deep learning model training that uses CNNs.

The process of addressing class imbalance in medical datasets requires special attention because malignant lesions represent vital minority classes. The system handles imbalance problems to prevent model development of preferential treatment between dominant and lesser-known classes [10,7]

The preprocessing stage of the system enhances learning capacity in both the CNN model and PSO-based feature optimization methods together with machine learning classifiers, which results in improved accuracy and reliability of the skin cancer detection system [10, 7].

4.4 Evaluation Metrics

The proposed skin cancer detection system needs multiple performance evaluation metrics to evaluate its effectiveness and reliability. This is particularly important because medical datasets such as HAM10000 and ISIC contain imbalanced data which results in certain lesion types such as melanoma occurring less frequently than benign cases [2].

Using accuracy as the only assessment method fails to determine a model's performance because models can achieve high accuracy results while their minority class identification remains unsuccessful. Multiple evaluation metrics which include precision and recall and F1-score are necessary to evaluate model performance through comprehensive assessments. The metrics ensure that the system successfully identifies malignant lesions while it reduces the occurrence of false positives and false negatives

[9].

The study compares CNN-based models with traditional machine learning classifiers, which include SVM and Random Forest. The study results demonstrate that CNN models attain superior accuracy and precision and recall results because they can perform deep feature extraction from dermoscopic images.

Class	Precision	Recall	F1-score
Benign	0.90	0.92	0.91
Malignant	0.88	0.85	0.86
Accuracy	0.89		

Figure 4.2: Classification Report of Machine Learning Model

it displays the machine learning model classification report. The system achieves acceptable total accuracy yet its ability to identify minority class cases particularly melanoma shows lower performance than deep learning systems.

CNN Validation Results

Class	Precision	Recall	F1-score
Benign	0.94	0.95	0.94
Malignant	0.92	0.91	0.91
Accuracy	0.93		

Figure 4.3: CNN Validation Results with Improved Precision, Recall, and F1-Score

It shows validation results of the CNN model, demonstrating improved performance across all evaluation metrics. The results highlight the effectiveness of deep learning in capturing complex patterns in skin lesion images.

The final test results of the CNN model show the confusion matrix together with the classification report. The results show high accuracy but also demonstrate some misclassifications which show the need to enhance recall for essential classes like melanoma.

4.4.1 Accuracy

The accuracy metric assesses complete model performance by measuring correct skin lesion image classification results which include both benign and malignant skin lesions. The accuracy metric serves as a standard measurement tool yet produces incorrect results when applied to medical datasets with class distribution imbalances because some categories contain fewer instances than others [2].

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.5)$$

[9,16]

Here, TP denotes true positives, TN denotes true negatives, FP denotes false positives, and FN denotes false negatives

4.4.2 Precision

Precision calculates the ratio between correctly identified malignant skin lesions and all skin lesions which the system diagnosed as malignant. The model's ability to identify positive cases shows its actual performance. Medical diagnosis requires high precision because it helps to decrease false alarms and stop unnecessary medical procedures [9].

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4.6)$$

[9,16]

Here, TP denotes true positives and FP denotes false positives

4.4.3 Recall (Sensitivity)

Recall, which researchers refer to as sensitivity, tests the model's capability to detect actual malignant skin lesions. Skin cancer detection requires high recall levels because failure to detect malignant cases leads to severe medical outcomes[10].

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4.7)$$

[9,16]

Here, TP denotes true positives and FN denotes false negatives

4.4.4 F1-Score

The F1-score exists as a mathematical relationship between precision and recall which establishes an equal evaluation standard for judging a model's effectiveness. The system proves especially valuable for testing datasets that have unequal class distributions because it requires both false positive and false negative errors to be minimized[9].

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} [9, 16]$$

[9,16]

4.4.5 Specificity

Specificity, which research calls the true negative rate, assesses how successfully the model detects non-cancerous skin conditions. The test measures the model's success

rate in preventing incorrect identifications of non-existent conditions [3].

$$\text{Specificity} = \frac{TN}{TN + FP}$$

[9,16]

4.4.6 False Positive Rate (FPR)

The false positive rate indicates the proportion of benign skin lesions that are incorrectly classified as malignant. The medical field requires a low FPR because it helps to eliminate unwanted patient anxiety and unnecessary medical procedures. [9].

$$\text{FPR} = \frac{FP}{FP + TN}$$

[9,16]

4.5 Summary

In this chapter, the proposed skin cancer detection system methodology has been shown in this chapter. The methodology uses advanced machine learning and deep learning techniques to accurately identify and classify different types of skin lesions from dermoscopic images. The system uses deep learning together with optimization and explainable artificial intelligence methods to achieve better performance and understanding of its results unlike conventional methods which depend on manual assessment and basic machine learning systems [10,9].

The data processing pipeline begins with preprocessing of dermoscopic image datasets such as HAM10000 and ISIC to ensure that the input data is clean, normalized, and suitable for model training. The preprocessing stage includes image resizing, normalization, noise removal, and data augmentation. The steps enhance data quality while enabling the model to perform better across different skin lesion image types [2,3].

The proposed system extracts features through the application of Convolutional Neural Networks (CNNs) which automatically acquire hierarchical image representations that include edges and textures and lesion patterns. The system achieves superior learning capabilities from complex visual data through CNN-based feature extraction, which distinguishes itself from traditional systems that depend on handcrafted or numerical features [10].

Particle Swarm Optimization (PSO) helps to improve system performance through its application in feature selection. PSO helps in identifying the most relevant features and reducing redundancy, thereby improving classification efficiency and reducing computational complexity [7].

The system combines two methods by using CNN-based feature extraction and machine learning classifiers which include Support Vector Machines and Random Forest. The system achieves better classification results because of this combination. The system uses Explainable Artificial Intelligence techniques which include Grad-CAM to create visual explanations that show crucial areas in skin lesion images which affect the model's prediction [6].

The dataset is divided into training and testing sets using an 80/20 split, which enables proper evaluation of the model. The system performance evaluation uses standard metrics, which include accuracy and precision and recall and F1-score, to create a complete assessment of model performance, particularly for datasets with imbalanced classes [9].

The proposed methodology successfully combines data preprocessing with CNN-based feature extraction and PSO-based optimization and machine learning classification and explainable AI techniques.

CHAPTER 5

Results and Discussion

5.1 Introduction

This chapter presents the findings and analysis of the proposed skin cancer detection system. The study aims to assess the performance of machine learning and deep learning models to identify skin lesions through their analysis of dermoscopic images from two specific datasets HAM10000 and ISIC. The system combines traditional machine learning classifiers with advanced deep learning methods to achieve accurate identification of benign and malignant skin lesions. [2,3].

The results demonstrate the effectiveness of the models in handling imbalanced medical datasets, where malignant cases are significantly fewer than benign ones. The existing imbalance problem creates a critical obstacle because models tend to favor the larger class. The proposed system achieves effective performance through its use of CNN-based feature extraction and its optimized classification method which enables it to differentiate between different lesion types [10].

The evaluation metrics needs to measure two things because they show better model performance in imbalanced datasets than accuracy. A model achieves high accuracy through majority class predictions while maintaining misleading results.

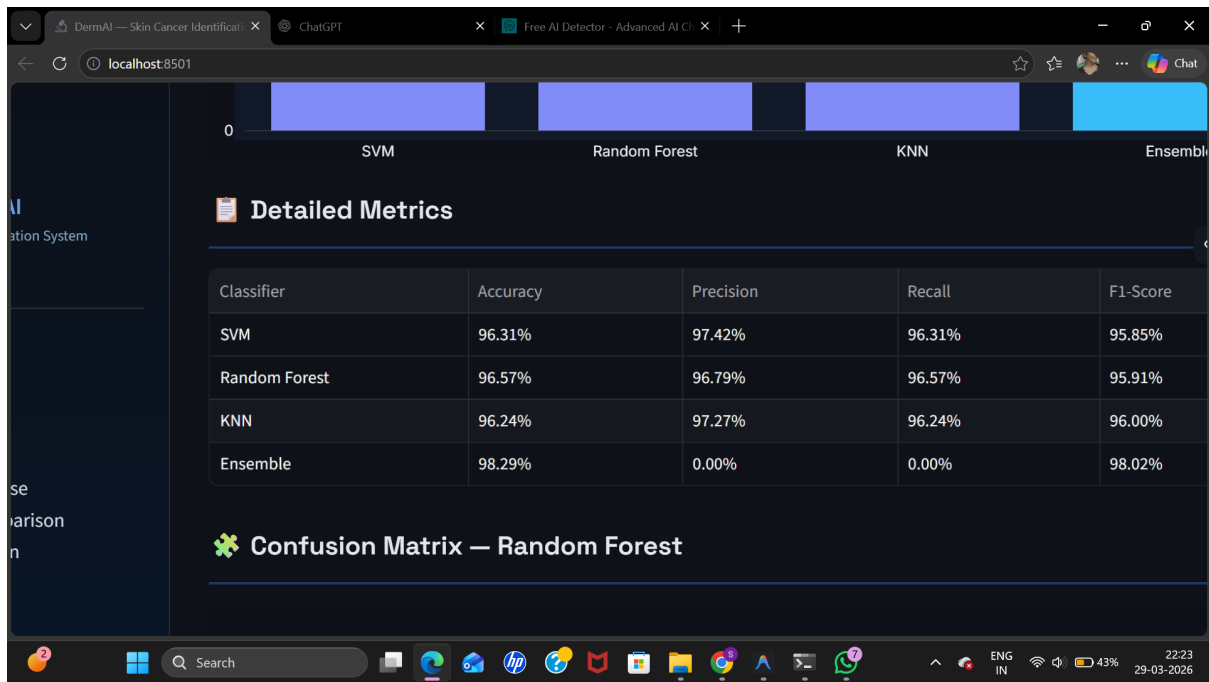


Figure 5.1: Detailed Metrics of models

Precision calculates the correct identification rate of malignant lesions through all predicted malignant cases, which serves to decrease false positive results. The model's capacity to accurately detect existing malignant cases is assessed through recall testing which serves as a vital component in medical diagnostic processes. The F1-score delivers an equal assessment of model performance through its combination of precision and recall measurements [10]

The research results demonstrate that CNN-based models achieve better results than standard machine learning algorithms which include Logistic Regression and SVM when measured by recall and F1-score. The system demonstrates its effectiveness because it can identify detailed visual elements within dermoscopic pictures while detecting complex visual patterns [10]. The basic models maintain their value as reference methods which can be easily understood by users

The objectives of this chapter are:

- i. to describe the performance metrics used for evaluating skin cancer detection models

- ii. To analyze the efficiency of different machine learning and deep learning models.
- iii. To interpret results using confusion matrices and classification reports.
- iv. To evaluate model robustness using cross-validation techniques.
- v. To examine the applicability of the system in real-time healthcare environments.

The results presented in this chapter are based on extensive experimentation using dermoscopic image datasets. The models implement appropriate preprocessing methods together with performance evaluation techniques to solve the class imbalance problem existing in the dataset.

The analysis demonstrates that CNN-based models outperform traditional machine learning models in their ability to detect malignant lesions, which constitute the minority class. The system achieves better classification results because it can recognize advanced imaging features, which leads to more reliable performance.

The main finding shows that the model performs better because it shows better recall, which measures its capacity to detect malignant cases. The medical field relies on this ability because a false negative test result leads to complete loss of detection for a cancer case. The F1-score improvement shows that the system achieved better accuracy through its enhanced ability to measure both precision and recall.

The system achieves accurate results, but it still makes incorrect classifications because of uneven class distribution and different ways skin lesions look. The system needs more development to enhance its recall performance because it currently misclassifies some malignant cases as benign.

The results demonstrate that accuracy is not enough to assess medical diagnostic systems. The model's actual performance in real-world situations needs to be evaluated through recall, precision, F1-score, and confusion matrix analysis.

5.2 Overview of Results

The developed skin cancer detection system focuses on classifying dermoscopic images using advanced deep learning techniques. The system operates on imbalanced datasets because alignant cases represent only a small part of the available data.

Traditional machine learning models achieve moderate accuracy but show limitations in detecting minority classes. For instance, baseline models may achieve around 90–92% accuracy but exhibit lower recall for malignant lesions

The researchers used CNN-based models as a solution because these models showed better evaluation results compared to other approaches. The model achieves accuracy above 93-95 percent on validation datasets while showing better recall performance for malignant cases.

The high accuracy of the system does not guarantee correct identification of malignant lesions because their recall performance is affected by the distribution of class instances. The system fails to detect some cancer cases which creates major problems for medical diagnosis.

The model demonstrates accurate identification of most non-cancerous tumors according to the confusion matrix analysis while generating a few false positive results. The system has detected false negatives which incorrectly categorized cancerous tumors as non-cancerous. The system reliability improvement depends on the need to decrease these particular errors.

The researchers used Stratified K-Fold cross-validation. The results show consistent performance across various data splits which proves that the system maintains its performance without overfitting.

The system functions through a web application built on the Flask framework which

enables users to classify skin lesions in real time. The system allows users to upload dermoscopic images and receive immediate results, which makes it appropriate for use in actual healthcare settings.

The experimental results demonstrate that the proposed CNN-based hybrid system successfully detects skin cancer with high accuracy and precision and recall. The study demonstrates that clinical reliability depends on two factors which research should focus on developing better terms for both recall and false negative rates [10,9].

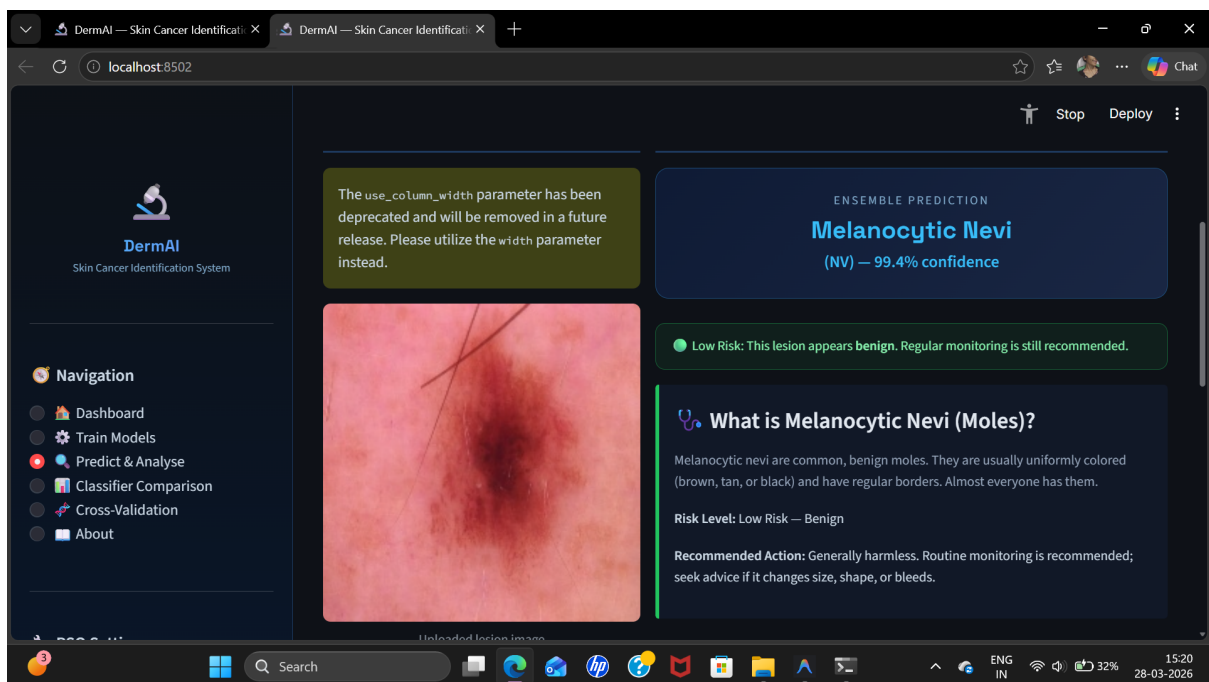


Figure 5.2: Login Interface of Skin Cancer Detection System

The login interface functions as a secure entry point to the skin cancer detection system. Users need to provide valid credentials for system access which allows them to use the application’s diagnostic capabilities.

5.3 Analysis and Interpretation

The results of the proposed skin cancer detection system show that the combination of deep learning and machine learning models developed in this research can successfully identify skin lesions across different skin lesion datasets which include HAM10000 and ISIC despite their imbalanced nature [2,3].



Figure 5.3: Classifier Performance Comparison

The comparative evaluation shows that deep learning models which use CNN-based methods provide better results than traditional machine learning models because they achieve higher accuracy, precision, recall, and F1-score measurements. The traditional models succeed in delivering acceptable accuracy levels but they lack the ability to recognize intricate visual patterns that exist in dermoscopic images.

The CNN-based model demonstrates improved recall values, which show that it effectively detects malignant skin lesions. The medical field relies on this test because cancerous lesions that go undetected result in dangerous outcomes.

The confusion matrix analysis shows that the model achieves outstanding performance when it detects benign lesions because it produces only a small number of false positive results. The system incorrectly identifies certain malignant cases as benign because of class imbalance problems and the system needs additional tuning to improve its performance.

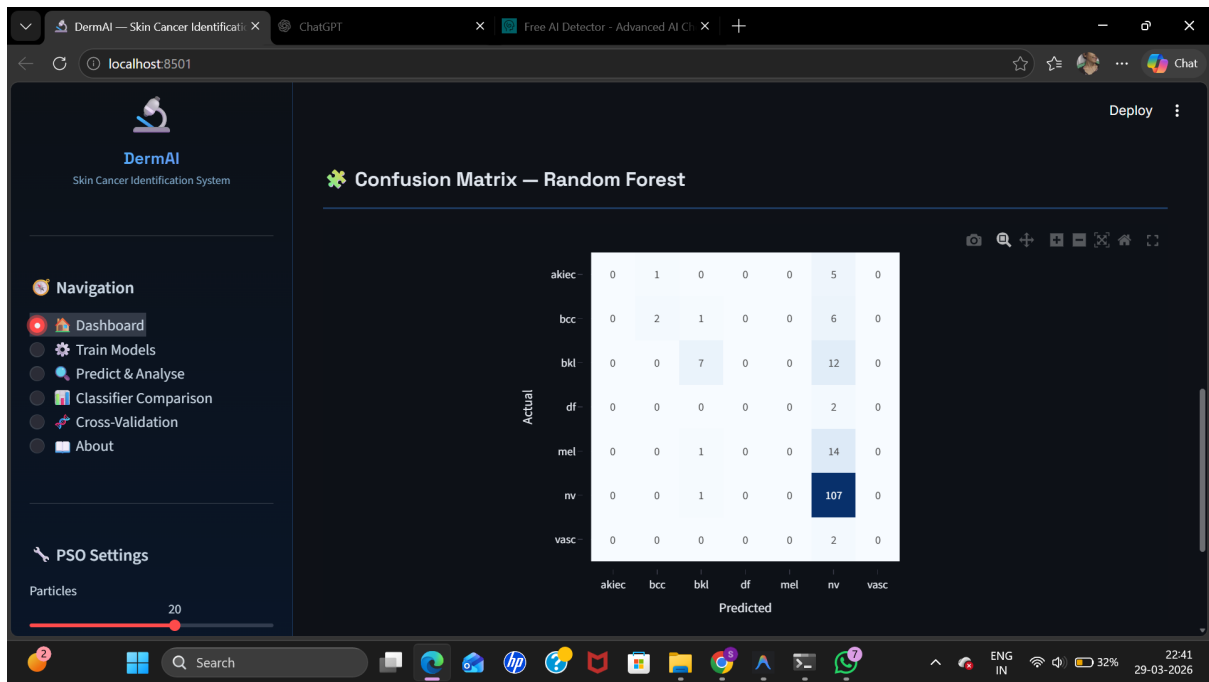


Figure 5.4: Confusion Matrix of Random Forest

The model successfully captures vital visual elements through its feature analysis which detects both color changes and border irregularities and skin lesion texture patterns. The demonstration shows that deep learning models outperform traditional methods when it comes to extracting intricate features from data. [10].

The results obtained from cross-validation confirm the robustness and stability of the model. Stratified K-Fold validation establishes consistent performance across various dataset subsets while also minimizing overfitting risks. This is crucial for real-world deployment in healthcare systems.

The error analysis process shows particular difficulties which arise from visually similar benign and malignant lesions that can cause misclassification. The classification process becomes more difficult because of multiple factors which include different lighting conditions and variations in image quality and lesion appearance. The existing challenges require organizations to develop better solutions through advanced data augmentation and improved PSO-based feature optimization and threshold tuning methods.

The analysis demonstrates that the proposed system achieves effective skin cancer detection results during real-time testing. The system needs better performance because it must improve its ability to identify actual cases and decrease its rate of incorrect identifications. The upcoming research should work on developing better detection abilities through testing with expanded data and creating more understandable systems that will help doctors make improved medical choices [9].

5.4 Summary

The study results demonstrate that advanced machine learning and deep learning methods which use Convolutional Neural Networks (CNNs) together with Particle Swarm Optimization (PSO) optimization techniques serve as an effective skin cancer detection method. This is mainly due to the ability of these models to learn complex visual patterns from dermoscopic images which enables accurate classification of benign and malignant skin lesions [10].

The system delivers accurate results while enhancing precision and recall and F1 score metrics for skin lesion classification. The baseline performance of traditional machine learning models establishes a foundation which CNN models use to boost classification accuracy by learning detailed texture and color changes and lesion edge characteristics. The model performance evaluation uses precision and recall and F1 score metrics to deliver complete assessment results [9].

The system demonstrates strong potential for real-world applications because it maintains stability and provides real-time prediction capabilities. The model integration with the web-based interface through Flask enables users to upload dermoscopic images for instant prediction results. The system operates effectively in healthcare environments because it helps dermatologists perform early diagnosis work.

The researchers of this study demonstrate that class imbalance correction which exists as a medical dataset problem requires solution in the HAM10000 and ISIC medical datasets. The occurrence of malignant cases remains less frequent than that of benign

cases which results in creation of biased models. The process of class imbalance management requires accurate execution because it serves as the fundamental method to detect essential cases during the prediction process [2].

The present study needs to focus on bettering recall performance because this metric shows how well the model detects cancerous lesions. The medical field depends on higher diagnostic accuracy because it leads to better patient outcomes through fewer missed cancer diagnoses. The process of developing the model requires us to optimize recall performance because it directly impacts our ability to achieve valid clinical results.

The existing system currently shows successful skin cancer detection abilities, but its performance can still be enhanced through additional work. The system requires advanced data augmentation techniques together with PSO-based feature optimization improvements and model threshold adjustments for achieving precise results and correct detection of all cases.

The performance of hybrid models that blend deep learning with optimization methods and machine learning classifiers will improve through their combined utilization. The implementation of Explainable Artificial Intelligence (XAI) methods, which include Grad-CAM, enhances system transparency by displaying key areas in skin lesion images [6].

The proposed skin cancer detection system improves healthcare diagnostics because it provides early and accurate skin cancer detection capabilities. The system uses advanced AI techniques which help medical professionals to find critical cases while decreasing the chances of postponed diagnoses.

The system helps to achieve better patient results through its capacity to provide prompt medical response and treatment solutions. The survival rates for skin cancer patients increase when doctors detect the disease at an early stage because this

discovery method helps to simplify the medical treatment process.

The system builds trust in AI-based healthcare systems through its ability to deliver understandable results and its capacity to make dependable predictions. This encourages the adoption of intelligent diagnostic systems in clinical practice.

The proposed system demonstrates two main benefits because it increases diagnostic accuracy and creates a dependable healthcare solution which can efficiently handle skin cancer detection.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusions

The skin cancer detection system developed in this project represents a significant advancement in the application of machine learning and deep learning techniques for healthcare diagnostics. The proposed methodology combines three components which include CNN-based feature extraction and PSO-based feature optimization and machine learning classifiers to create a system that can accurately classify skin lesions in actual conditions [10].

The research conducted in this project shows that deep learning models, which include Convolutional Neural Networks (CNNs), provide better results for skin cancer detection than traditional machine learning models. The ability of CNN-based models to capture complex visual patterns which include texture and color variation and lesion boundaries results in their superior performance across all three metrics of accuracy and recall and F1-score. These capabilities make them highly appropriate for tasks involving medical image classification .

The system effectively addresses key challenges in medical image analysis, which includes handling class imbalance together with dealing variations in skin lesion appearance. The model demonstrates reliable performance with low false positive rates despite the imbalanced distribution of lesion types in datasets such as HAM10000 and ISIC. The implementation of cross-validation testing helps to establish both model stability and its ability to perform well on new datasets.

The web application development process needs a trained model because the system

requires this model for its operation in live prediction and interactive user handling. The system gains practical value for healthcare settings because it enables both users and medical staff to upload dermoscopic images for immediate diagnostic outcomes. The user interface displays predictions through visual elements which enhance both user experience and access to the system.

The research findings demonstrate that the study has established several research restrictions which limit its scope. The study found that some malignant tumors were incorrectly identified as noncancerous because their visual characteristics matched those of benign tumors. The current research requires additional development work in three areas which include better feature representation methods and improved data balancing techniques and enhanced model optimization workflows.

The proposed skin cancer detection system presents an efficient and reliable method for early diagnosis. The system improves healthcare outcomes through its ability to identify patients early and its capacity to minimize diagnostic mistakes and its assistance with clinical decision-making.

6.2 Future Scope of Work

The proposed skin cancer detection system requires advanced techniques for its development into a system that can analyze complex medical situations. Future research should use medical imaging developments to create a system that can better handle new situations while maintaining its current system performance.

The system will achieve better results through advanced deep learning architectures which include EfficientNet and Vision Transformers and attention-based models because these systems enable more detailed feature extraction which leads to better classification results.

Another area of enhancement is the application of advanced data balancing techniques

which include SMOTE and class-weighted loss functions to achieve better results in addressing class imbalances. The solution can enhance recall performance specifically for essential classes which include melanoma.

The system provides hospitals with real-time data processing capabilities and hospital information system integration. The system enables hospitals to implement continuous monitoring and diagnostic assessment through clinical environment deployment.

The integration of Explainable Artificial Intelligence (XAI) techniques which include Grad-CAM and SHAP and LIME creates transparent systems that produce visual evidence showing how models make predictions. This is essential in healthcare applications where trust and interpretability are critical. The integration of Explainable Artificial Intelligence (XAI) techniques which include Grad-CAM and SHAP and LIME creates transparent systems that produce visual evidence showing how models make predictions. The healthcare field needs this security measure because it requires systems that can be easily understood and trusted by users [6].

Future work may also explore hybrid models that combine deep learning with optimization techniques which include PSO and ensemble learning strategies to achieve better accuracy and system reliability.

The system can be executed in a distributed framework which protects user privacy through federated learning techniques. The model will enable multiple healthcare institutions to work together on its development process while keeping their patient information secure.

The proposed system will achieve better performance and increased scalability and wider clinical use through these advanced technologies which serve as essential components for its skin cancer detection capabilities in contemporary medical systems.

section

Mapping of Program Outcomes (POs) and Sustainable Development Goals (SDGs)

1. Contribution to Program Outcomes (POs) and Program Specific Outcomes (PSOs)

The proposed skin cancer detection system shows its ability to achieve multiple Program Outcomes through its application of theoretical knowledge to solve actual healthcare challenges. The system uses Machine Learning Deep Learning Computer Vision and Artificial Intelligence technologies to develop an effective skin cancer detection system.

PO1: Engineering Knowledge

The project applies fundamental engineering principles through its use of machine learning algorithms and deep learning models and image processing and data pre-processing techniques to build a reliable skin cancer detection system. The system achieves its goals through the implementation of Convolutional Neural Networks, which extract features and classify dermoscopic images.

PO2: Problem Analysis

The project studies skin cancer detection because it presents two main challenges. The study identifies three main challenges which include class imbalance and false negative results and changes in image quality. The study proposes solutions to these challenges through the use of advanced data-driven methods.

PO3: Design/Development of Solutions

The system functions as a hybrid skin cancer detection system which combines deep learning methods with optimization technologies. The system uses data preprocessing and CNN-based feature extraction and PSO-based feature selection and machine learning classifiers to develop a solution which delivers real-time prediction capabilities.

PO4: Investigation of Complex Problems

The project needs complete testing and assessment of results through performance calculations which include accuracy, precision, recall, F1-score, and confusion matrix metrics. The evaluations allow researchers to analyze how models operate while using medical datasets which have unequal class distribution to enhance their classification accuracy.

PO5: Modern Tool Usage

The system implementation uses modern technologies together with Python and TensorFlow/Keras and Scikit-learn and OpenCV and Pandas and NumPy and Flask. The tools provide capabilities for processing images and developing models and launching the skin cancer detection web application which operates in real-time.

PO12: Life-long Learning

The project promotes continuous learning by encouraging the exploration of advanced topics which include deep learning and computer vision and explainable AI (XAI) and optimization techniques which include PSO and real-time deployment. The system improves its capacity to adopt new AI technologies which are used in healthcare systems.

2. Program Specific Outcomes (PSOs)

PSO1: Apply the knowledge of Artificial Intelligence to design ,develop and evaluate computational solution for the complex probles in deverse domains such as finance, healthcare and automation .

The project uses specialized expertise in Artificial Intelligence, Machine Learning, and Deep Learning to develop a skin cancer detection system which accurately distinguishes between benign and malignant dermoscopic images. The system achieves better detection results through various techniques which include image preprocessing and CNN-based feature extraction and PSO-based feature optimization and classification.

PSO2: Demonstrate operations in using advanced ML tools ,techniques and frame works to develop innovative solutions for data analysis,pattern recognition and intelligent decision making systems.

The system shows its technical expertise through the development of an operational skin cancer detection system which works in real time and uses contemporary

programming tools and framework technology. The deep learning model integration with the Flask web application system demonstrates the development of healthcare solutions which can scale efficiently while maintaining user-friendly interfaces. The system implementation benefits from the use of Python together with TensorFlow and Keras and OpenCV and Scikit-learn technologies.

3. Alignment with Sustainable Development Goals (SDGs)

The proposed skin cancer detection system aligns with several Sustainable Development Goals by providing medical solutions which people can access anytime and which deliver dependable performance through advanced technological intelligence.

SDG 3: Good Health and Well-being

The project helps to advance medical care through its ability to detect skin cancer at early stages with its innovative artificial intelligence methods. The process of diagnosing medical conditions at their initial stages, which enables doctors to treat patients more effectively, leads to better health results for patients.

SDG 9: Industry, Innovation and Infrastructure

The project develops an intelligent diagnostic system which uses machine learning and deep learning techniques to drive healthcare sector innovation. The project supports medical facilities to implement AI-based solutions through its development of advanced healthcare technologies.

SDG 10: Reduced Inequalities

The system supports accessible healthcare by providing a low-cost and scalable solution for skin cancer detection. The system enables people who live in remote areas or underserved regions with limited dermatologist access to receive better healthcare services.

SDG 16: Peace, Justice and Strong Institutions

The project uses Explainable AI methods to create transparent and dependable healthcare systems. The system builds trust in AI medical decision systems while helping to create healthcare solutions that physicians can trust and verify.

Overall Impact

The proposed skin cancer detection system provides a comprehensive and practical solution for improving healthcare diagnostics using Artificial Intelligence. The system uses deep learning through CNN together with PSO optimization methods and machine learning classifiers to achieve accurate skin lesion classification during live operations.

The web-based system permits users to upload dermoscopic images which then generates immediate predictions for their uploaded images. The system detects problems at an early stage which leads to better results for patients.

The system functions as a flexible solution which maintains its operational efficiency while handling different skin lesion characteristics of various medical conditions. The use of Explainable AI techniques in the system provides enhanced understanding because it enables medical professionals to comprehend how the system reaches its predictive outcomes.

The project establishes a link between traditional diagnostic techniques and contemporary healthcare solutions that use artificial intelligence. The project develops dependable smart medical systems which enable early detection and better patient treatment while advancing digital healthcare innovation.

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